

Temperature Converter (Celsius ↔ Fahrenheit) Program

1. Prompt

Create a Python program that converts temperature values between Celsius and Fahrenheit. The program should:

- Ask the user to choose the conversion type (Celsius to Fahrenheit or Fahrenheit to Celsius).
- Ask the user to enter the temperature value.
- Apply the correct conversion formula.
- Display the converted temperature.

This program demonstrates basic input handling, conditional statements, and mathematical calculations in Python.

2. Steps / Algorithm

Step 1: Start the program.

Step 2: Display menu options for conversion:

1. Celsius to Fahrenheit
2. Fahrenheit to Celsius

Step 3: Ask the user to choose an option.

Step 4: Input the temperature value.

Step 5: If option = 1, apply formula:

$$\text{Fahrenheit} = (\text{Celsius} \times 9/5) + 32$$

Step 6: If option = 2, apply formula:

$$\text{Celsius} = (\text{Fahrenheit} - 32) \times 5/9$$

Step 7: Display the converted temperature.

Step 8: End the program.

3. Code (Suitable for Prompt Engineering)

```
print("Temperature Converter")
print("1. Celsius to Fahrenheit")
print("2. Fahrenheit to Celsius")
```

```
choice = int(input("Enter your choice (1 or 2): "))
```

```
if choice == 1:
```

```
    celsius = float(input("Enter temperature in Celsius: "))
    fahrenheit = (celsius * 9/5) + 32
    print("Temperature in Fahrenheit:", fahrenheit)
```

```
elif choice == 2:
```

```
    fahrenheit = float(input("Enter temperature in Fahrenheit: "))
```

```
celsius = (fahrenheit - 32) * 5/9
print("Temperature in Celsius:", celsius)
```

else:

```
print("Invalid choice")
```

4. Debugging Method / Code

Common Errors and Solutions:

1. ValueError

Occurs when the user enters text instead of a number.

Solution: Enter numeric values only.

2. Invalid Choice

If the user enters a number other than 1 or 2.

Solution: Display an error message and ask again.

Debugging Example:

try:

```
value = float(input("Enter temperature: "))
print("Valid input")
```

except:

```
print("Invalid input. Please enter a number.")
```

5. Output

Example 1:

Temperature Converter

1. Celsius to Fahrenheit

2. Fahrenheit to Celsius

Enter your choice (1 or 2): 1

Enter temperature in Celsius: 25

Temperature in Fahrenheit: 77.0

Example 2:

Temperature Converter

1. Celsius to Fahrenheit

2. Fahrenheit to Celsius

Enter your choice (1 or 2): 2

Enter temperature in Fahrenheit: 98.6

Temperature in Celsius: 37.0